VaultSign AI

Project Antigravity: Local, Zero-Trust Contract Review for Freelancers

Final Implementation Plan (v2). Built with open-weight models for the open-source AI "Side Hustle" challenge.

1. Executive Overview

Independent contractors and freelancers regularly receive jargon-heavy client contracts that may contain predatory clauses. They often cannot afford legal counsel for every gig. Uploading these documents to a cloud AI tool creates real risk, because many contracts carry confidentiality terms that restrict sharing with third parties (the "NDA paradox" from v1, now stated more carefully).

VaultSign AI runs the analysis **on the user's own machine** using Ollama and an open-weight language model. It reads a contract, flags risky clauses **with the exact quoted text and page reference**, explains them in plain English, and drafts polite counter-offer emails. In the default local mode, the document is processed on the user's machine and is not sent to any external service.

> **One-line pitch:** a contract red-flag reviewer that works with your Wi-Fi off, so the contract never has to leave your laptop. It is a review aid, not a lawyer.

2. What Changed from v1

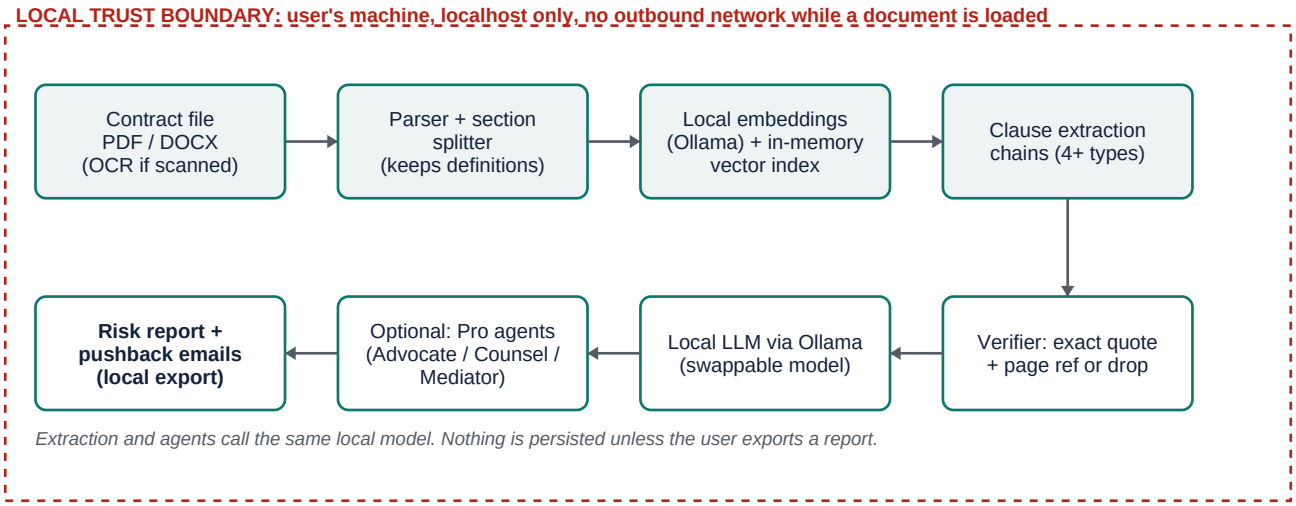
This plan keeps the core idea and stack of the original proposal. The changes below address the weaknesses found in the review. Items marked **Kept** are carried over as written in v1; **Changed** and **New** are improvements.

Area	v1 claim	v2 decision	Status
NDA argument	Cloud upload "directly" violates NDAs	"May violate / creates real risk"; depends on clause wording	Changed
Privacy wording	"Air-gapped", "100% data privacy", "not a single byte leaves"	"Offline-by-design local mode" with a test anyone can repeat; no 100% claim	Changed
Cloud tier	$3/contract private cloud, still described as private	Moved out of MVP; later option with an honest, different trust model	Changed
Model	Llama 3 (8B)	Swappable; default chosen from current Ollama library by benchmark, long context required	Changed
Accuracy	"Highly capable"	Measured on a labeled test set; quote + page proof for every flag	New
Multi-agent	Core differentiator	Kept, but as a Pro extra only if it beats single-pass in a measured test	Changed
UI	Streamlit/Gradio and Electron both mentioned	Streamlit or Gradio on localhost for MVP; desktop packaging later	Changed
Clause types	4 types	Same 4 plus 4 more freelancer traps	Kept + New
Inputs	PDF	PDF and DOCX; OCR for scans as optional	New
Legal	No disclaimer	Prominent not-legal-advice notice; jurisdiction warning	New
Market	60M+ audience, $15,000 MRR Y1	Unsourced figure removed; targets shown as assumptions with the math	Changed
Stack: RAG, Ollama, CrewAI, Mediator email generator	As specified	Retained	Kept

3. Core Technology Stack

Layer	Choice	Role and notes
Inference engine	Ollama (kept from v1)	Local execution of open-weight models. Lets users swap models to fit their hardware. Telemetry-free operation is verified, not assumed (see Section 5).
Core LLM	Open-weight model, default picked by benchmark	v1 named Llama 3 (8B). Original Llama 3 8B has a short context window, which is a poor fit for 50-page contracts. Shortlist 2 to 3 current long-context models from the Ollama library (Llama 3.1 8B or newer, plus one or two alternatives) and keep the best on your test set. Verify current availability yourself before locking in.
Small-hardware fallback	A smaller model (about 3B or less)	For low-RAM machines, with a visible quality warning in the UI.
Document parsing	pdfplumber / pypdf; python-docx	Text PDFs and DOCX. Optional OCR (pytesseract) for scans.
RAG pipeline	LangChain (kept from v1)	Splits by contract section, not fixed size, so definitions and cross-references survive. Local embeddings through Ollama. In-memory vector index, nothing written to disk.
Agents (Pro)	CrewAI (kept from v1)	Advocate, Corporate Counsel, Mediator, all running on the local model.
UI	Streamlit or Gradio on localhost	Bound to 127.0.0.1 only. Pick one for MVP. Analytics and telemetry explicitly disabled.

4. System Architecture and Data Flow



The new **Verifier** step is the main safety mechanism. A flagged clause is shown only if the model's quoted text is found verbatim in the parsed contract. Quotes that cannot be matched are discarded, which removes fabricated clauses from the report.

5. Privacy and Trust Model

v1 promised 100% privacy and an air-gapped system. A tool that blocks its own network calls is offline-by-design, not air-gapped, and the claim should be one a skeptic can verify. v2 therefore commits to the following.

- **Local-only by default:** UI bound to 127.0.0.1; no cloud API keys anywhere in the code path.
- **No-network mode:** while a document is loaded, the app refuses outbound requests (application level). For stronger assurance the docs explain how to run it with Wi-Fi off or with an OS firewall rule.
- **Telemetry off:** UI framework analytics disabled and documented; dependency list kept short and reviewed.
- **Verification demo:** run an analysis with the network disconnected, and optionally capture traffic to show zero outbound connections. This is shown in the challenge post.

- **No persistence:** vector index lives in memory only; reports are saved only when the user exports them to a local file.
- **Honest limits:** the OS, other software on the machine, and the one-time model download are outside the app's control. The model is downloaded *before* any contract is opened.

6. Implementation Phases

Phase names from v1 are kept. The MVP is deliberately small, as the challenge asks for "small, specific, and useful". Durations are planning estimates, not commitments; scale them to your real deadline.

Phase	Goal	Deliverables	Done when
1. Zero-Trust Foundation (kept, tightened)	Working offline pipeline	Install script or clear setup guide for Ollama and dependencies; localhost UI; network-block mode; telemetry disabled; PDF and DOCX parsing	App analyzes a sample contract with Wi-Fi off
2. "Red Line" Translation Engine (kept, extended)	Reliable flags with proof	Section-aware RAG; extraction chains for all clause types in Section 7; Verifier; plain-English summaries; local report export	Every flag shows quote + page; unmatched quotes are dropped
2b. Evaluation (new)	Know the real accuracy	Labeled test set; precision/recall per clause; model comparison (2 to 3 models); single-pass baseline	Results table ready for the post
3. Pushback Generator (kept)	Usable counter-offers	Email template per flagged clause; polite, firm tone; editable in UI	Net-90 example produces a ready-to-send email
4. Pro Simulator (kept, now conditional)	Agent debate	Advocate, Corporate Counsel, Mediator flow; ablation vs single-pass	Included only if it catches more real issues, or users clearly prefer its output
5. Packaging and Launch (new)	Ship it	README, demo video, offline demo, challenge post, license	Post published

7. Clause Detection Specification

The four clause families from v1 are kept. Four more common freelancer traps are added. Each extraction must return: clause type, **verbatim quote**, page/section, risk level, plain-English explanation, and a suggested ask.

#	Clause family	What to look for	Origin
1	IP transfer / work for hire	Perpetual or irrevocable assignment; assignment before payment; no carve-out for pre-existing tools	v1
2	Extended payment terms	Net-60 or Net-90; payment conditional on client approval or on client being paid	v1
3	Broad non-compete	Wide scope, long duration, no geography limit, restricts other clients	v1
4	Asymmetric liability / indemnification	One-sided indemnity; uncapped liability; freelancer bears all risk	v1
5	Unlimited revisions / scope creep	"Until satisfied" language; vague deliverables; no change-order process	New
6	Termination and kill fee	Termination at will with no payment for work done; no kill fee	New
7	Late-payment consequences	No late fee or interest; payment withheld for disputes	New
8	Exclusivity	Restrictions on working for other clients during or after the term	New

> **Output rule:** if nothing is found for a clause family, the report says "Not detected. Please verify manually." It never says "no problems." Small models can miss clauses, so absence of a flag is not proof of safety.

8. Evaluation Plan (New)

v1 asserted that the model is highly capable but did not measure it. v2 treats accuracy as a deliverable.

- **Test set:** 20 to 30 contracts, mixing public sample agreements and synthetic ones you write, with the target clauses labeled by hand. Check the license of any public dataset before use (for example, a contract-review benchmark such as CUAD); never use a real client's confidential contract.
- **Metrics per clause family:** precision (how many flags are correct), recall (how many real clauses are caught), and false-fabrication rate (quotes that fail the Verifier).
- **Model comparison:** run the same set on 2 to 3 open-weight models plus the small fallback; report accuracy against speed on your hardware.
- **Multi-agent ablation:** single-pass vs. Advocate/Counsel/Mediator on the same set. Record recall gain, added false positives, and added time.
- **Report honestly:** publish numbers even if modest. A tested tool with known limits is more credible than an untested claim.

Clause family	Model A	Model B	Small fallback	Single-pass	Multi-agent
IP / work for hire	TBD	TBD	TBD	TBD	TBD
Payment terms	TBD	TBD	TBD	TBD	TBD
Non-compete	TBD	TBD	TBD	TBD	TBD
Liability / indemnity	TBD	TBD	TBD	TBD	TBD

Results table template. Fill in with measured recall/precision. TBD means not yet measured.

9. Multi-Agent Negotiation Simulator (Pro)

The three roles from v1 are kept. The change is how they are used: they are constrained to the contract text and are evaluated against the single-pass baseline before being sold as a differentiator.

Agent	Objective (from v1)	v2 guardrail
The Advocate	Identify any clause that limits the freelancer's future earning potential or delays compensation.	Must quote the clause it criticizes.
The Corporate Counsel	Defend the contract as written; justify clauses as standard practice.	Must not invent legal precedent or statistics; claims about "standard practice" are labeled as arguments, not facts.
The Mediator	Synthesize the debate and generate a compromise clause.	Output goes through the same Verifier for any quoted text; compromise wording is marked as a suggestion to review with a lawyer.

Fallback: if the ablation shows little benefit, replace the three-agent debate with a single "devil's advocate" pass, which gives most of the value at far lower compute.

10. Pushback Email Generator

Kept from v1: after analysis, the tool produces polite, professional correspondence rather than raw legal advice. The v1 example is retained:

> "To align with my standard operational cash flow, I kindly request adjusting the payment terms from Net-90 to Net-30."

- One editable template per flagged clause; tone options (friendly, neutral, firm).
- The user's own name, project, and clause reference are filled in locally.
- Each email includes an editable placeholder for a justification, so it is not a generic copy-paste.

11. Legal and Safety Guardrails (New)

- **Not legal advice:** show a clear notice on first run, on every report, and in the README. VaultSign AI is a review aid that helps users spot issues and prepare questions.
- **Jurisdiction warning:** enforceability of non-competes, IP assignment, and liability limits varies by country and region. The report reminds users to check local rules.
- **No guarantees:** do not claim the tool prevents harm or replaces counsel. Avoid the phrase "legal co-pilot" in marketing; prefer "contract red-flag reviewer".
- **Before charging money:** have a qualified lawyer review the terms of service, disclaimer, and any wording that could be read as legal advice. This plan is not a substitute for that review.

12. Go-to-Market and Monetization (Revised)

The tier structure from v1 is kept for the local product. The cloud tier is deferred, and all numbers are presented as assumptions to test rather than facts.

Tier	v1 plan	v2 plan
Basic Core (Free)	Open-source app; basic RAG clause extraction on own hardware with Ollama	Kept. Clause extraction with quotes and plain-English explanation. This is the challenge submission.
Pro Desktop License ($10/month)	Unlocks CrewAI simulator, Pushback Email Generator, Predatory Clause Database updates	Kept as the starting hypothesis. Also test a one-time license or per-contract credits, since many freelancers sign only a few contracts a year.
Cloud-Secure Web ($3/contract)	Private cloud with open-weight models	Deferred to after launch. If built, described honestly: the file does leave the user's machine, so use encryption in transit, in-memory processing, immediate deletion, and open-source server code. Not marketed as "zero-trust local". Compute cost per contract must be measured first.

Open-core note: because the product is open source, anyone could fork it and unlock Pro features. Decide the license up front. Common options are an open core with paid modules under a different license, or selling convenience, updates, and the curated Predatory Clause Database rather than the code itself. Choose before launch.

Predatory Clause Database: the only ongoing-value feature in v1. Define who curates it and how: start with 30 to 50 clause patterns with plain-English explanations, reviewed by a lawyer, and update on a regular schedule.

Revenue math (assumption, not a forecast)

Item	Figure	Note
v1 target MRR (Year 1)	$15,000	From v1
Subscribers needed at $10/month	1,500	$15,000 divided by $10; Pro-only, ignores churn and fees
Audience claim in v1 (60M+)	Unverified	Removed until a source is found; do not quote in the post without one
Required next step	Acquisition plan	Name 3 to 4 channels (freelancer forums, communities, newsletters) and set a realistic early goal, such as the first 20 paying users

13. Why Open Innovation Matters Here (for the Challenge Post)

The challenge asks where an open approach worked better than a closed one. VaultSign AI's answers, each with proof to include:

Argument	Why open wins	Proof to show
Privacy	The contract is processed on the user's machine, so there is no third-party server to trust or to breach confidentiality terms.	Offline demo with Wi-Fi off; optional zero-traffic capture
Cost	50-page PDFs are token-heavy. A paid API charges per token; local inference costs only the user's electricity.	Cost comparison for one 50-page contract, using current published API prices you verify yourself
Swap and tune	Models can be changed to fit hardware or accuracy, and prompts or fine-tuning can target freelancer clauses.	Side-by-side model results from Section 8
No lock-in	No dependence on one provider's pricing, policy changes, or availability.	Same pipeline running on 2 to 3 different models
Product viability	v1 notes that proprietary APIs would break the user's NDA and add high token costs. Both are fixed by local open weights.	Retained from v1

14. Risk Register

Risk	Impact	Mitigation
Small model misses or invents clauses	High	Verifier (quote + page), "Not detected, verify" wording, measured evaluation
Slow on low-end laptops	Medium	State minimum specs, show typical timings, small-model fallback
Privacy claims challenged	High	Accurate wording, reproducible offline test, short dependency list
Legal liability / unauthorized-practice concerns	High	Disclaimers, jurisdiction note, lawyer review before charging
Scope too large for deadline	Medium	MVP limited to Phases 1, 2, 2b and 3; agents and cloud come later
Open-source forks undercut Pro tier	Medium	Decide license early; sell updates, database, and convenience
Scanned or messy contracts	Medium	Optional OCR; clear warning when text extraction is poor

15. Challenge Submission Checklist

	Item
[ ]	Working local app: upload PDF/DOCX, flagged clauses with quotes and page numbers
[ ]	Offline demo (video or GIF) with Wi-Fi disabled
[ ]	Evaluation table with measured results and model comparison
[ ]	Pushback email generated for at least one real-looking example (Net-90 to Net-30)
[ ]	Not-legal-advice notice visible in the app and README
[ ]	Post section explaining why open innovation mattered (Section 13)
[ ]	Clear license and README with setup steps and minimum hardware
[ ]	Any market figure in the post has a source; otherwise it is left out

Note: this plan is a project plan, not legal or financial advice. Model names, versions, library features, and prices change quickly; verify them at build time.